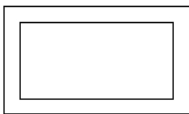
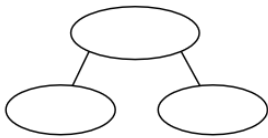
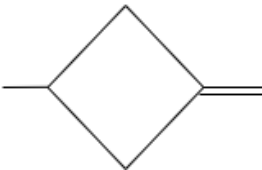
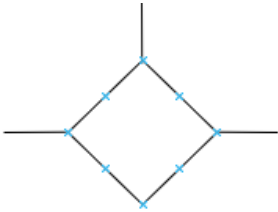
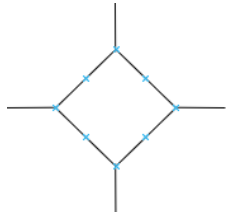
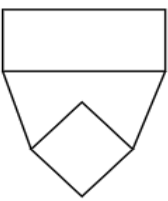
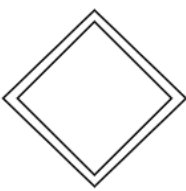
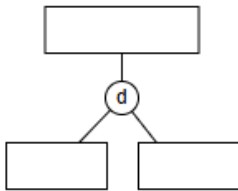
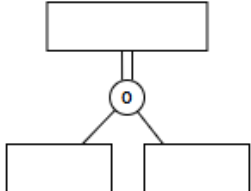


CSE370 Final Review

Final Syllabus:

- ER/EER → Things to keep in mind:

Strong entity 	Weak entity 	Simple Attributes 	Partial Attribute 
Composite Attribute 	Multivalued Attribute 	Composite + Multivalued 	Key Attribute 
Derived Attribute 	Binary Relationship 	Ternary Relationship 	N-ary Relationship(N>3) 
Recursive Relationship 	Identifying Relationship 	Disjoint Specialization  Disjoint ⇒ d Optional Participation ⇒	Overlapping Specialization  Overlapping ⇒ o Total Participation ⇒
Cardinality Ratio 1:1 ⇒ One to One, 1:N ⇒ One to Many, M:N ⇒ Many to Many [Must Use different letters for M:N]		Existence Dependency Single Line for Optional Participation Double Line for Total Participation NOTE: If Cardinality and Participation are not Clearly mentioned OR can't be inferred from Question, Make Logical Assumption	

1. Schema Mapping

Step	Mapping Process	Details
Step 1	Mapping of Regular/Strong Entity Types	Primary Key should be underlined
Step 2	Mapping of Weak Entity Types	Both Partial Key and Foreign Key of Strong Entity should be underlined
Step 3	Mapping of Binary 1:1 Relationship Types	- FK placed in the relation with Total Participation, otherwise any relation is fine. - FK points towards referenced PK (not underlined, use an arrowhead)
Step 4	Mapping of Binary 1:N Relationship Types	- FK placed in the relation on the N side. - FK points towards referenced PK (not underlined, use an arrowhead)
Step 5	Mapping of Binary M:N Relationship Types	- Create a separate relation with both participating entities' Keys as a composite key. - Both keys should be underlined, and both are FKs (they point to their respective PKs).
Step 6	Mapping of Multivalued Attributes	- Create a separate relation with the primary key of that entity and the multivalued attribute. - Both the key and the attribute should be underlined. FK points towards PK.
Step 7	Mapping of N-ary Relationship Types	- Create a separate relation where the PK will contain all the participating entities' keys. - All keys should be underlined and FKs point toward PK.

Superclass / Subclass schema mapping

8A: Can be applied for all types (total or optional, disjoint or overlapping)

8B: Can be applied if the specialization is **Total Only**

8C: Can be applied if the specialization is **Disjoint Only**

8D: Can be applied for all types (total or optional, disjoint or overlapping)

- Option 8A: Multiple relations-Superclass and subclasses
- Option 8B: Multiple relations-Subclass relations only
- Option 8C: Single relation with one type attribute
- Option 8D: Single relation with multiple type attributes

SCHEMA MAPPING EXAMPLES [Pictures Attached, Link Below]

- [Binary Relationships](#)
- [Specialization/ Generalization](#)
- [Multivalued](#)
- [Recursive M:N](#)

2. **Normalization**

- Informal guidelines and Functional Dependencies (FD)
- 1NF disallows
 - composite attributes
 - multivalued attributes
 - nested relations
- 2NF
 - Must be in 1NF,
 - All attributes depend on the whole key. Disallows Partial Dependencies (transitive dependencies can exist in 2NF)
- 3NF
 - Must be in 2NF
 - All attributes depend on nothing but the key (No transitive dependencies exist)
- [Normalization Example Solved](#)

3. **Indexing**

- Primary index (clustering) VS secondary index (non-clustering)
- Sparse VS Dense index
- Multi-level indexing
- [B+ Tree Insertion Simulation](#) [[Online Simulator](#)]
- [B+ Tree Deletion](#) (Slide)
- **B+ Tree Update:** First Delete Old value, Then Insert New value

- Hashing → Has storage units called buckets. It uses a hash function to locate entries for access. Buckets store entries with pointers to records
- Bucket Overflow and Why It Occurs.
- Overflow chaining and open addressing are used to resolve overflow.
- [Practice Examples](#)

4. [Transaction Management](#)

- **ACID** (Atomicity, Consistency, Isolation, Durability) Properties
- Recovery Subsystem → Why is it needed?
- Concurrency Control → Why is it needed?
- Lost Update Problem, Temporary Update Problem, Incorrect Summary Problem, Non-repeatable Read, Phantom Read.
- All Four Isolation levels.
- [Practice Examples](#)

5. **SQL (Same as LAB)**

Select

From

Where

Group by

Having

Order by

Like %, _

Aggregate functions (min, max, count, sum, avg)

Subquery, Correlated Subquery

Inner join (Very important)

Some Question Types:

- Given the Schema, asked to write some queries
- Given a SQL query, asked to identify errors
- Given a SQL query, asked to write an alternate solution that gives same result

PRACTICE SHEET LINK: [8. CSE370_Mysql \[Practice Sheet\]](#)